SELF-INITIATED PROJECT · NIBM SRI LANKA · 2026

From an empty VM to a campus lab platform.

NIBM had no student lab server. On my own initiative I built one from scratch on the campus ESXi host, then grew it into a secure, self-service platform: 30+ open-source services, one identity, zero manual account juggling.

SH

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swipe →

01 · VERSION 1: BUILT WHILE LEARNING

It worked.
It wasn't safe yet.

I built the first version myself while still learning: services first, guard rails later. Auditing my own work showed what an institution needs:

Vector database without auth
Anyone on the network could read or wipe every collection.

Every port exposed
No host firewall, including the Docker-socket admin panel (that's root on the server).

No backups, no alerts
A storage mount failed silently for two weeks before I caught it.

Default passwords, scattered accounts
Default admin logins still active; every app had its own unrelated user list.

02 · ARCHITECTURE

Everything goes through one front door.

Campus users
LAN or ZeroTier mesh

AdGuard DNS
split-horizon *.lab names

Reverse proxy
OpenResty + Lua guards, HTTPS only

Authentik: one identity
self-registration with approval · groups · forward-auth · per-user API keys minted at login

Dashboards
admin cockpit / student hub, same URL

Apps
Forum · Git · Docs · Vault · n8n · Qdrant · Jellyfin · PDF & dev tools

Admin consoles
Portainer · DNS · Proxy · game server, behind SSO

Watchers
Uptime Kuma · Beszel · Dozzle · Diun

Alerts
ntfy push to the admin's phone

Backups
restic nightly → Synology NAS

Ubuntu 24.04 on a Dell PowerEdge (ESXi). 34 containers, 16 GB RAM, all defined as code.

Your index number is your login. Everywhere.

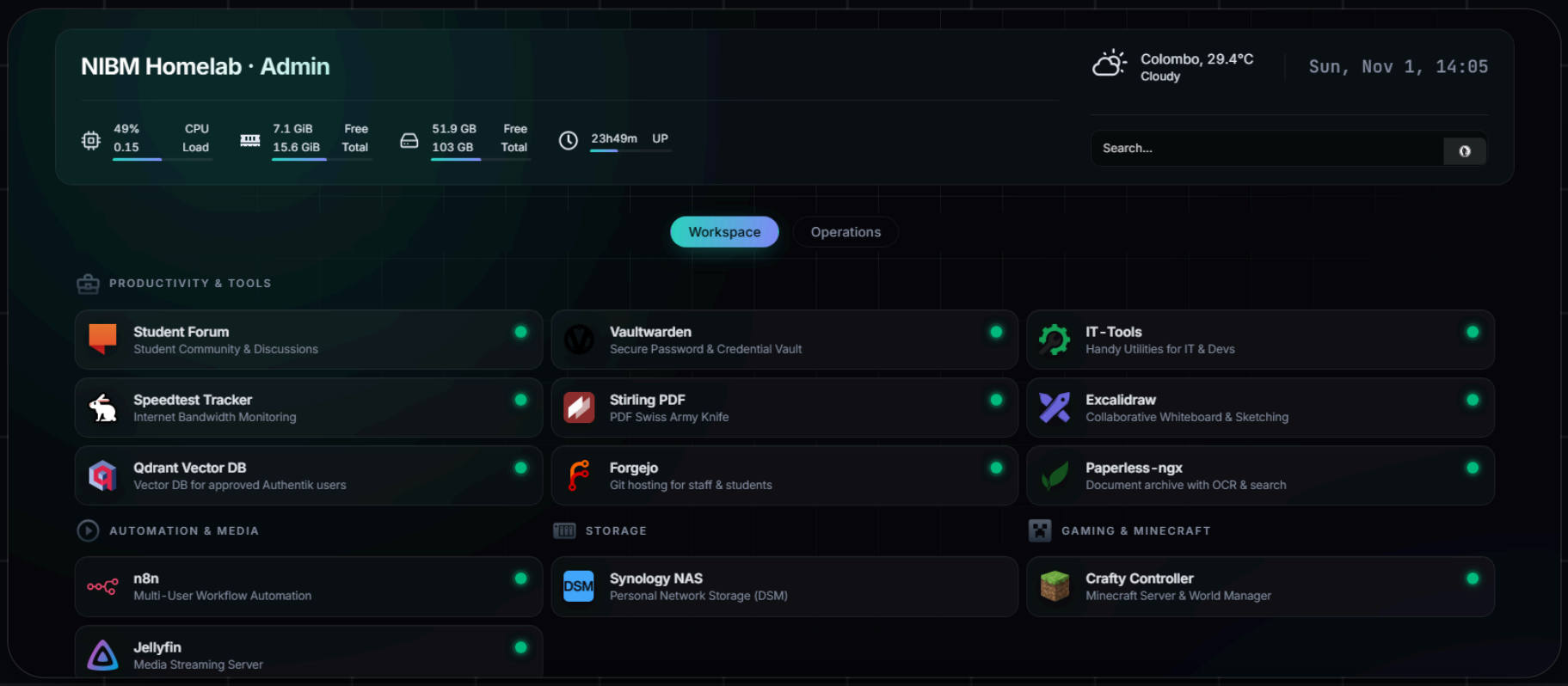
1. **Self-registration, admin-approved**
   Students sign up with their index number (format enforced, e.g. cohndne251f-019). Accounts stay inactive until approved, and the admin gets a push alert.

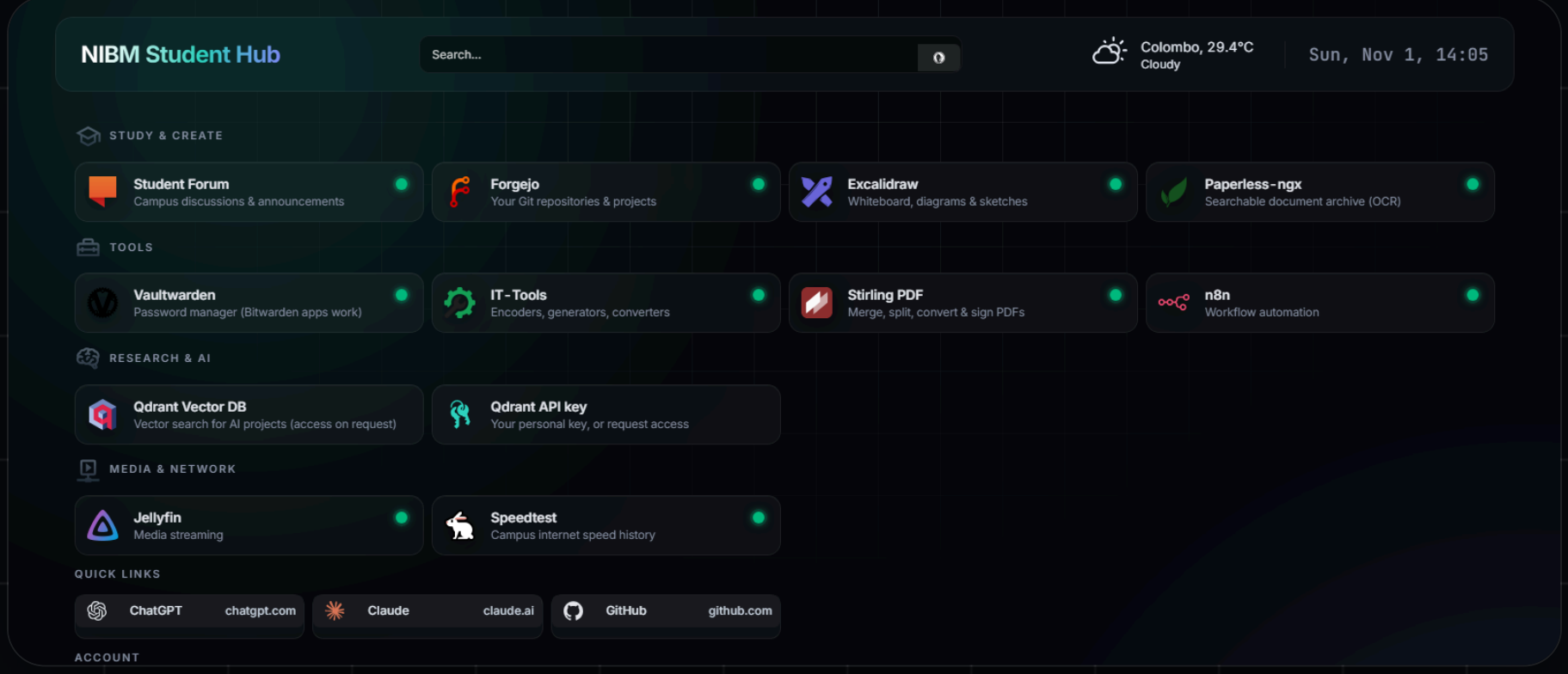
2. **Groups decide everything**
   Dashboards, consoles and database permissions follow group membership. No per-app user juggling.

3. **Identity-bound app accounts**
   Apps with their own logins only accept sign-ups that match the signed-in lab identity. A Lua guard in the proxy enforces it.

04 · ROLE-BASED DASHBOARDS

Same URL. Different view.





Admins get the full cockpit; students get a clean hub. The proxy picks the instance from the identity provider's group header.

05 · SECURE BY DEFAULT

Least privilege, enforced.

- ✓ **Scoped, expiring API keys**
  Each login mints a personal vector-DB token: read-only, per-collection or manage. The master key is never exposed.
- ✓ **Two locks on every console**
  SSO gate for admins first, then the app's own login.
- ✓ **Firewall that understands Docker**
  Published container ports filtered in DOCKER-USER; the LAN reaches only web, DNS, SSH and git.
- ✓ **Hardened host**
  Key-only SSH, fail2ban, pinned images, secrets out of config, 22 audit findings fixed and logged.

06 · AUTOMATED LIFECYCLE

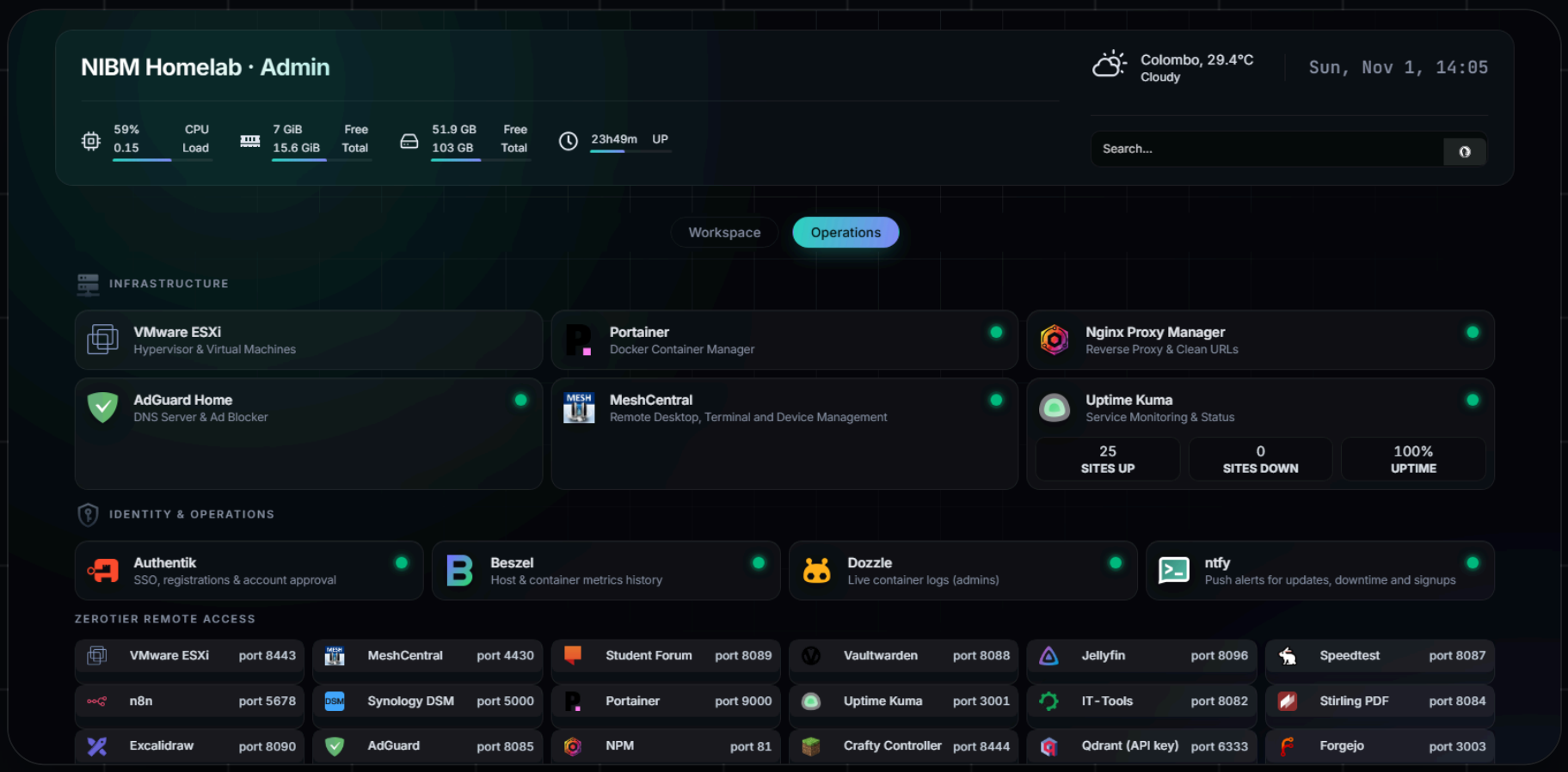
Onboarding to offboarding, hands-off.

1. Student registers with their index number
2. Admin approves in one click (push alert)
3. Dashboard, forum and Git unlock automatically
4. Vector-DB access via self-service request
5. Account disabled: access ends everywhere within 2 min

A sync job keeps app accounts in step with the identity provider. Leavers lose access without anyone touching five admin panels.

07 · OPERATIONS

If it breaks, the phone knows first.



25 uptime probes · metrics · live logs

Downtime, failed backups, sign-ups and new versions all push to ntfy.

Nightly encrypted backups, restore-tested

Consistent DB dumps + app data to the NAS, 7/4/6 retention, weekly integrity checks.

The numbers.

34
containers, all pinned and defined as code

24
HTTPS hosts generated from one script

22
security findings fixed or tracked, incl. 3 critical

−22%
RAM in use, while adding 13 services

3
end-to-end test suites with real logins

1
identity per person, across every service

09 · STACK

Built entirely on open source.

VMware ESXi · Ubuntu 24.04 · Docker Compose · Authentik · Nginx Proxy Manager · OpenResty · Lua · AdGuard Home · ZeroTier · Cloudflare Tunnel · Qdrant · Forgejo · Flarum · Paperless-ngx · Vaultwarden · n8n · Uptime Kuma · Beszel · Dozzle · Diun · ntfy · restic · iptables · fail2ban

Documented for the next person.

Admin guide (navigation to handover), student guide and a from-scratch setup guide, so the lab I started keeps serving students after I leave.

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Questions and feedback welcome